# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building **AI-enabled full-stack applications**, **secure APIs**, and **cloud workflows**. Strong in Python, React, SQL, automated testing, and healthcare-oriented product systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – June 2026*
- Architected **Azure SQL and Blob services** for 500+ users, tuning indexed queries to cut dashboard latency from **1.2 seconds to 350 milliseconds**.
- Built **fault-tolerant job execution** for 1,000+ monthly workflows with retries and isolation, improving throughput by **3x** under concurrent production workloads.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Engineered **AI-powered Flask APIs** and React workflows handling **10,000+ daily requests**, integrating multimodal grading and secure ingestion for production educational operations.
- Designed **multithreaded processing** for 1,000+ uploads on AWS RDS-backed services, reducing feedback turnaround by **60%** while improving throughput and debugging.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built **transformer evaluation pipelines** across **100K+ trajectory logs**, enabling reproducible experimentation and high-confidence benchmarking for long-horizon model behavior.
- Architected **spatial-temporal encoders** with map-aware attention, improving closed-loop planning stability by **15%** while deepening ML systems and failure-analysis rigor.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered **Python API and ETL microservices** with SQL, AWS Glue, and S3, improving response times by **30%** across distributed data workflows.
- Maintained **Jenkins CI pipelines** and 20+ automated suites with PyTest and Mockito, increasing release reliability by **68%** across shared environments.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Built **MERN APIs and WebSocket modules** supporting **2M+ synchronized records**, improving consistency across real-time modular workflows.
- Integrated **Jenkins CI/CD**, Docker releases, and monitoring-driven delivery practices, cutting release time by **50%** while improving deployment consistency.

## Skills

**Languages**: **Python**, **TypeScript**, JavaScript, SQL, Java, C++, Bash; **Web**: React, HTML5, CSS3, ES6+, Flask, FastAPI, REST APIs, WebSockets, microservices
**Data & Cloud**: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Azure, Azure SQL, AWS RDS, S3, Blob Storage, AWS Glue, ETL, indexing, query tuning
**AI & Quality**: GPT-4o, Claude, OCR, PyTorch, Jenkins, CI/CD, Git, Docker, PyTest, Mockito, automated testing, logging, monitoring, Sentry, Agile/Scrum

## Projects

**HMIS Live** Live | *MERN, ABDM APIs, TypeScript, Twilio, Sentry*
- Built a **healthcare workflow platform** with ABDM APIs, Twilio alerts, and Sentry monitoring, enabling consent-based patient-history access and reducing administrative workload by **30%**.

**Linguistix** | *Python, AWS, Docker, CI/CD*
- Developed a containerized **Python REST API** on AWS with automated CI/CD and validation checks, achieving **87% accuracy** while supporting reliable data-processing workflows.

**Chat2SQL** Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** for **10+ databases**, accelerating query authoring and schema exploration through unified Java and TypeScript services.